Circuit Fundamentals Teaching Guide

For NLC STEM Club UAV Electrical Team

From Basic Concepts to Real System Calculations

Electronics & Power Team

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Purpose

This guide teaches the electrical theory and math we used to design the UAV power system. All examples are from OUR ACTUAL PROJECT.

Target Audience: Beginners with little/no electrical experience

Teaching Approach:

1. Learn the fundamental concept
2. See the formula
3. Apply it to our UAV system
4. Try a practice problem

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1 The Big Three (Voltage, Current, Resistance)

Think of electricity like water flowing through pipes...

VOLTAGE (V)

- **Measured in:** Volts (V)
- **Water analogy:** Water pressure in the pipe
- **Symbol:** V or E
- **Example:** Our battery is 14.8V (pushes electrons with 14.8V of "pressure")

CURRENT (I)

- **Measured in:** Amperes or Amps (A), milliamps (mA)
- **Water analogy:** How much water flows per second
- **Symbol:** I
- **Example:** Our flight controller uses 250mA = 0.25A (250 milliamps of electrons flowing through it)
- **Conversion:** 1000 mA = 1 A

RESISTANCE (R)

- **Measured in:** Ohms (Ω)
- **Water analogy:** Narrow pipe restricts water flow
- **Symbol:** R
- **Example:** A resistor that limits current flow

2 Ohm's Law (The Most Important Formula)

Ohm's Law states the relationship between Voltage, Current, and Resistance:

$$V = I \times R$$

Where:

- V = Voltage (Volts)
- I = Current (Amps)

- R = Resistance (Ohms)

2.1 Rearranged Forms

$$I = \frac{V}{R} \quad \text{(Find current if you know voltage and resistance)}$$
$$R = \frac{V}{I} \quad \text{(Find resistance if you know voltage and current)}$$

2.2 Memory Trick - Ohm's Law Triangle

Cover what you want to find, and you see the formula!

- Cover V: You see $I \times R \rightarrow V = I \times R$
- Cover I: You see $V/R \rightarrow I = V/R$
- Cover R: You see $V/I \rightarrow R = V/I$

2.3 Example from Our UAV

Servo Resistance Calculation

Question: If our servo draws 180mA when moving, and it's powered by 5V, what is its effective resistance?
Given:

- V = 5V
- I = 180mA = 0.18A (convert to amps!)

Find R:

$$R = \frac{V}{I}$$
$$R = \frac{5\text{V}}{0.18\text{A}}$$
$$R = 27.78\ \Omega$$

Answer: The servo has an effective resistance of ∼28Ω when moving.

2.4 Practice Problem 1

Our GPS module uses 60mA at 5V. What is its resistance?
Given: V = _____, I = _____
Find: R =?
$R = V/I$ = _____ = _____
Answer: _____ Ohms

3 Electrical Power (How Much Energy?)

Power measures how fast electrical energy is used or converted.

$$P = V \times I$$

Where:

- P = Power (Watts)
- V = Voltage (Volts)
- I = Current (Amps)

3.1 Other Forms (Using Ohm's Law)

$$P = I^2 \times R$$
$$P = \frac{V^2}{R}$$

3.2 Why We Care

- Power tells us how much energy a component uses
- Higher power = more battery drain
- Higher power = more heat generation
- We need to ensure our BECs can supply enough power

3.3 Example from Our UAV - Motor Power

Motor Power at 50% Throttle

Question: At 50% throttle, our motor draws 10A from the 14.8V battery. How much power is it using?
Given:

- V = 14.8V
- I = 10A

Find P:

$$P = V \times I$$
$$P = 14.8\text{V} \times 10\text{A}$$
$$P = 148 \text{ Watts}$$

Answer: The motor uses 148W at 50% throttle.

3.4 Example from Our UAV - BEC Capacity

BEC Maximum Power

Question: Our ESC has a 5V 3A BEC. What is the maximum power it can supply?
Given:

- V = 5V
- I = 3A (max)

Find P:

$$P = V \times I$$
$$P = 5\text{V} \times 3\text{A}$$
$$P = 15 \text{ Watts}$$

Answer: The BEC can supply up to 15W of power.

This is WHY we calculated our system needs!

- Flight controller: 5V × 0.25A = 1.25W
- GPS: 5V × 0.06A = 0.3W
- Receiver: 5V × 0.09A = 0.45W
- Servos (all moving): 5V × 0.72A = 3.6W
- Arduino + Sensors: 5V × 0.263A = 1.32W
- **TOTAL = 6.92W**

But 15W > 6.92W, so we're safe, right?
WRONG! Peak servo current can exceed 3A → BEC overload → WHY WE NEED DUAL-BEC!

3.5 Practice Problem 2

Our External UBEC supplies 5V at 6A. How many watts can it provide?
Given: V = _____, I = _____
Find: P =?
$P = V \times I$ = _____ = _____
Answer: _____ Watts

4 Battery Capacity (How Long Will It Last?)

Battery capacity tells us how much charge a battery can store.
Measured in: Amp-Hours (Ah) or milliamp-hours (mAh)

- 1 Ah = 1000 mAh

- Our battery: 2200 mAh = 2.2 Ah

Meaning: The battery can supply 2200mA for 1 hour OR 1100mA for 2 hours OR 4400mA for 0.5 hours (30 minutes)

$$\textbf{Runtime (hours)} = \frac{\textbf{Capacity (Ah)}}{\textbf{Current Draw (A)}}$$

4.1 Example from Our UAV - Flight Time Calculation

Flight Time at Cruise Power

Question: Our motor draws 10A at cruise. Our battery is 2200mAh. How long can we fly?
Given:

- Battery Capacity = 2200 mAh = 2.2 Ah
- Current Draw = 10A

Find Runtime:

$$\begin{aligned}
\text{Runtime} &= \frac{\text{Capacity}}{\text{Current}} \\
\text{Runtime} &= \frac{2.2 \text{ Ah}}{10\text{A}} \\
\text{Runtime} &= 0.22 \text{ hours} = 13.2 \text{ minutes}
\end{aligned}$$

BUT WAIT! We should NEVER fully drain a LiPo battery!
Safe limit: Use only 80% of capacity
Usable capacity: 2200mAh $\times$ 0.8 = 1760 mAh = 1.76 Ah

$$\begin{aligned}
\text{Safe Runtime} &= \frac{1.76 \text{ Ah}}{10\text{A}} \\
&= 0.176 \text{ hours} = 10.6 \text{ minutes}
\end{aligned}$$

Answer: Safe flight time is about 10-11 minutes at cruise power.

4.2 Converting Units

Our 5V system draws 1383mA typical. How long can it run on the battery?

Step 1: Convert mA to A
1383 mA $\div$ 1000 = 1.383 A

Step 2: BUT the battery is 14.8V, and the 5V system is after conversion! We need to account for efficiency losses ($\sim$85% efficient)

Power at 5V: $P = 5\text{V} \times 1.383\text{A} = 6.92\text{W}$
Current from 14.8V battery: $I = 6.92\text{W}/14.8\text{V} = 0.468\text{A}$

Step 3: Calculate runtime
Runtime = 1.76 Ah/0.468A = 3.76 hours

This is why the 5V electronics are NOT the limiting factor - the motor is!

4.3 Practice Problem 3

Our Arduino system draws 263mA at 5V. If we had a separate 2200mAh battery just for it, how long would it last? (Assume you can use 80% of capacity)

Given: Capacity = 2200mAh × 0.8 = _____ mAh
Current = _____ mA

Runtime = Capacity / Current = _____ hours

Answer: _____ hours = _____ minutes

5 Series vs Parallel Circuits

5.1 Series: Components in a Single Line

SERIES Configuration

Components in a single line (one path for current)

- **Current** is THE SAME through all components
- **Voltage** DIVIDES across components
- **Total Resistance:** $R_{\text{total}} = R_1 + R_2 + R_3$

Example: Battery cells in series (4S LiPo)

- Each cell: 3.7V
- 4 in series: 3.7V + 3.7V + 3.7V + 3.7V = 14.8V

5.2 Parallel: Components on Separate Branches

PARALLEL Configuration

Components on separate branches (multiple paths)

- **Voltage** is THE SAME across all components
- **Current** DIVIDES among branches
- **Total Resistance:** $\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

Example: All our 5V components connected to BEC

- Each device gets 5V
- Total current = FC + GPS + RX + Arduino + Servos

5.3 Our UAV Power System is Parallel

```
5V BEC → Flight Controller (250mA)
(Rail 1) → GPS Module (60mA)
```

```
        → ELRS Receiver (90mA)
        → Arduino System (263mA)

Total Current = 250 + 60 + 90 + 263 = 663 mA
```

ALL devices get 5V (parallel voltage rule)
TOTAL current is the SUM (parallel current rule)

5.4 Practice Problem 4

Rail 2 powers 4 servos in parallel. Each servo draws 180mA when moving. What is the total current draw from Rail 2?

Given: 4 servos, each 180mA
Total Current = _____ + _____ + _____ + _____ = _____ mA
Answer: _____ mA = _____ A

6 Wire Gauge & Current Capacity

Wires have resistance! Thicker wire = less resistance = can carry more current.

Wire sizes are measured in AWG (American Wire Gauge):

- SMALLER number = THICKER wire = MORE current capacity
- LARGER number = THINNER wire = LESS current capacity

6.1 Wire Gauge Reference Table

AWG	Diameter	Max Current	Our Use
12	2.05mm	25-30A	(too thick)
14	1.63mm	15-20A	Battery → ESC/UBEC
16	1.29mm	10-13A	Motor wires
18	1.02mm	7-10A	5V Power Distribution
20	0.81mm	5A	Servo power
22	0.64mm	3A	Signal wires
24	0.51mm	2A	Sensor I2C
26	0.40mm	1A	UART/SPI

Table 1: Wire Gauge Current Capacity

6.2 Why This Matters

If wire is too thin for the current:

- Wire heats up ($I^2 \times R$ power dissipation)
- Voltage drop occurs ($V_{\text{drop}} = I \times R_{\text{wire}}$)
- Can cause fire!

If wire is too thick:

- Wastes money
- Adds unnecessary weight to UAV
- Harder to work with (stiff, bulky)

6.3 Example from Our UAV - Battery to ESC Wire

Wire Selection for High Current

Question: Our motor can draw up to 30A burst. What wire gauge do we need from battery to ESC?
Given:

- Max Current = 30A

From table:

- 14 AWG can handle 15-20A (not enough!)
- 12 AWG can handle 25-30A (just barely enough)

Recommendation: Use 12 AWG or 14 AWG with short run (Our ESC comes with XT60 and appropriate wire pre-attached)
Answer: 12-14 AWG wire is appropriate.

6.4 Practice Problem 5

Our External UBEC supplies up to 6A to Rail 2 (servos). What is the minimum wire gauge we should use?

Given: Max Current = _____ A
From table: Look up wire gauge that can handle _____ A
Answer: Use at least _____ AWG wire.

7 Voltage Regulation & BECs

BEC = Battery Elimination Circuit

- Takes high voltage input (14.8V from battery)
- Outputs stable low voltage (5V for electronics)
- Allows us to use ONE battery for everything

7.1 Linear Regulator (Not Used in Our System)

Linear Regulator

14.8V → [Linear Reg] → 5V

- Simple, cheap
- Wastes energy as heat: $P_{\text{wasted}} = (V_{\text{in}} - V_{\text{out}}) \times I$
- Example: $(14.8\text{V} - 5\text{V}) \times 3\text{A} = 29.4\text{W}$ wasted as heat!
- Efficiency: ∼40-60%

7.2 Switching Regulator / UBEC (What We Use!)

Switching Regulator / UBEC

14.8V → [Switching Reg] → 5V

- More complex, more expensive
- Efficient energy conversion
- Efficiency: ∼85-95%
- Minimal heat generation

7.3 Power Conservation

In an ideal world: $P_{\text{in}} = P_{\text{out}}$ (energy is conserved)

$$P_{\text{in}} = V_{\text{in}} \times I_{\text{in}}$$
$$P_{\text{out}} = V_{\text{out}} \times I_{\text{out}}$$

For switching regulator (assume 85% efficient):
$P_{\text{out}} = P_{\text{in}} \times 0.85$

7.4 Example from Our UAV - BEC Current Calculation

BEC Input Current

Question: Our 5V system draws 3A. How much current does the BEC draw from the 14.8V battery? (Assume 85% efficiency)
Step 1: Calculate output power

$$P_{\text{out}} = V_{\text{out}} \times I_{\text{out}}$$
$$P_{\text{out}} = 5\text{V} \times 3\text{A} = 15\text{W}$$

Step 2: Calculate input power (accounting for losses)

$$P_{\text{in}} = \frac{P_{\text{out}}}{\text{efficiency}}$$
$$P_{\text{in}} = \frac{15\text{W}}{0.85} = 17.65\text{W}$$

Step 3: Calculate input current

$$I_{\text{in}} = \frac{P_{\text{in}}}{V_{\text{in}}}$$
$$I_{\text{in}} = \frac{17.65\text{W}}{14.8\text{V}} = 1.19\text{A}$$

Answer: The BEC draws 1.19A from the 14.8V battery to supply 3A at 5V.
KEY INSIGHT: High voltage → Low current (that's why we use 14.8V battery!)

7.5 Practice Problem 6

Our External UBEC supplies 5V @ 6A to the servos. How much current does it draw from the 14.8V battery at full load? (Assume 85% efficiency)

Step 1: P_{out} = _____ V × _____ A = _____ W
Step 2: P_{in} = _____ W / 0.85 = _____ W
Step 3: I_{in} = _____ W / 14.8V = _____ A
Answer: _____ A from battery

8 Real System Design - Why Dual-BEC?

Now let's put it all together with OUR ACTUAL DESIGN DECISION.

8.1 Single-BEC Configuration (Original Plan - REJECTED)

```
14.8V Battery


40A ESC with 5V 3A BEC

    → Motor (14.8V)
```

```
→ 5V Rail (3A max)

      → Flight Controller (250mA)
      → GPS (60mA)
      → Receiver (90mA)
      → Arduino + Sensors (263mA)
      → 4× Servos (720mA moving, 1200mA peak!)
```

PROBLEM CALCULATION:

- Typical: $250 + 60 + 90 + 263 + 720 = 1383$ mA [OK] Under 3A limit
- Peak: $250 + 60 + 90 + 263 + 1200 = 1863$ mA [OK] Still under 3A

But what if ONE servo stalls (jammed control surface)?

- One stalled servo: 650mA
- Three moving servos: $3 \times 300\text{mA} = 900\text{mA}$
- Other electronics: $250 + 60 + 90 + 263 = 663\text{mA}$
- **TOTAL** $= 650 + 900 + 663 = 2213$ mA [OK] Still under 3A!

SO WHY DID WE REJECT THIS?

1. NO SAFETY MARGIN - operating at 60-75% of max continuously
2. BEC THERMAL STRESS - constant high current = heat = premature failure
3. VOLTAGE SAG - under heavy load, 5V drops to 4.8V → brownouts
4. SINGLE POINT OF FAILURE - if BEC fails, you lose EVERYTHING

8.2 Dual-BEC Configuration (Our Design - APPROVED)

```
14.8V Battery


XT60 Y-Harness Splitter

   → 40A ESC with 5V 3A BEC

         → Motor (14.8V)

         → Rail 1: Flight-Critical (3A available)

                → Flight Controller (250mA)
                → GPS (60mA)
                → Receiver (90mA)
                → Arduino + Sensors (263mA)
                      TOTAL: 663 mA (22% of 3A) [SAFE]
```

```
→ External 5V 6A UBEC

      → Rail 2: Servo Power (6A available)

            → 4× Servos (720mA typ, 1200mA peak)
                  PEAK: 1200 mA (20% of 6A) [SAFE]
```

ADVANTAGES:

OK Flight-critical systems isolated from servo current spikes

OK Each rail has 3× safety margin

OK If servo BEC fails, flight controller still powered → controlled landing

OK No voltage sag on critical systems

OK Professional aerospace-grade redundancy

8.3 The Math That Proved We Needed Dual-BEC

ESC BEC Capacity: 5V × 3A = 15W available

- Flight Critical Load: 5V × 0.663A = 3.3W (22% utilization) [OK]
- Servo Load (peak): 5V × 1.2A = 6W (40% utilization at peak) [OK]

Single-BEC Peak: 3.3W + 6W = 9.3W (62% utilization) - Marginal!
Dual-BEC Peak: Rail 1 = 3.3W (22%), Rail 2 = 6W (20%) - Excellent!

Cost Analysis:

- Cost of External UBEC: $12-15
- Cost of Dual-BEC Architecture: $12-15 total (just the UBEC!)
- Cost of crashing due to BEC failure: $500+ (entire project)

DECISION: $12 insurance policy → APPROVED!

9 Practice Problems - Full System Calculations

9.1 Problem Set A: Current Calculations

1. The ELRS receiver uses 90mA. If powered by 5V, how much power does it use?
$P = V \times I =$ _____ × _____ = _____ W

2. Our motor draws 18A at 75% throttle from 14.8V. How much power?
$P =$ _____ × _____ = _____ W

3. Three components draw 100mA, 250mA, and 80mA in parallel. What is total?
$I_{\text{total}} =$ _____ + _____ + _____ = _____ mA

9.2 Problem Set B: Battery Life

4. Battery: 2200mAh. Motor draws 14A. Flight time? (Use 80% of capacity)
Usable capacity: 2200mAh × 0.8 = _____ mAh = _____ Ah
Runtime = _____ Ah / _____ A = _____ hours = _____ minutes

5. If we reduce motor current to 8A, what's the new flight time?
Runtime = _____ Ah / _____ A = _____ hours = _____ minutes

9.3 Problem Set C: System Design

6. You add an FPV camera (200mA) to Rail 1. What's the new total current?
Original Rail 1: 663mA
With camera: 663mA + _____mA = _____ mA
Is this under the 3A limit? _____

7. Design check: Rail 2 has 4 servos. Each can stall at 650mA. If all four stall simultaneously (worst case), what current is drawn?
I_{total} = 4× _____ mA = _____ mA = _____ A
UBEC capacity: 6A
Margin: _____ A - _____ A = _____ A remaining
Is this safe? _____

9.4 Problem Set D: Efficiency & Conversion

8. BEC converts 14.8V to 5V at 85% efficiency. If output is 5V @ 2A, what power is drawn from the battery?
P_{out} = _____ V × _____ A = _____ W
P_{in} = _____ W / 0.85 = _____ W
I_{in} = _____ W / 14.8V = _____ A

9. If the BEC were only 70% efficient (linear regulator), how much power would be wasted as heat?
P_{in} = 10W/0.70 = _____ W
P_{wasted} = _____ W - 10W = _____ W of heat!

10 Answer Key

10.1 Practice Problems

Practice Problem 1: GPS Resistance
$R = V/I$ = 5V/0.06A = 83.33 Ω

Practice Problem 2: UBEC Power
$P = V \times I$ = 5V × 6A = 30 Watts

Practice Problem 3: Arduino Runtime
Capacity = 2200mAh × 0.8 = 1760 mAh
Runtime = 1760 mAh/263 mA = 6.69 hours ≈ 6 hours 41 minutes

Practice Problem 4: Servo Total Current
Total = 180 + 180 + 180 + 180 = 720 mA = 0.72 A

Practice Problem 5: Wire Gauge for 6A
Use at least 18 AWG wire (rated 7-10A)

Practice Problem 6: UBEC Battery Current
$P_{\text{out}} = 5\text{V} \times 6\text{A} = 30\text{W}$
$P_{\text{in}} = 30\text{W}/0.85 = 35.3\text{W}$
$I_{\text{in}} = 35.3\text{W}/14.8\text{V} = 2.38$ A

10.2 Problem Sets

Problem Set A:

1. $P = 5\text{V} \times 0.09\text{A} = 0.45\text{W}$
2. $P = 14.8\text{V} \times 18\text{A} = 266.4\text{W}$
3. $I_{\text{total}} = 100 + 250 + 80 = 430\text{mA}$

Problem Set B:

4. 1.76 Ah/14A = 0.126 hours = 7.5 minutes
5. 1.76 Ah/8A = 0.22 hours = 13.2 minutes

Problem Set C:

6. 663mA + 200mA = 863mA (Yes, under 3A)
7. $4 \times 650\text{mA} = 2600\text{mA} = 2.6\text{A}$; Margin: $6\text{A} - 2.6\text{A} = 3.4\text{A}$ (Yes, safe)

Problem Set D:

8. $P_{\text{out}} = 10\text{W}$; $P_{\text{in}} = 11.76\text{W}$; $I_{\text{in}} = 0.795\text{A}$
9. $P_{\text{in}} = 14.29\text{W}$; $P_{\text{wasted}} = 4.29\text{W}$

11 Teaching Tips

11.1 For In-Person Sessions

1. Start with water analogy - everyone understands water pressure/flow
2. Use the Ohm's Law triangle - visual memory is powerful
3. Always use OUR UAV examples - makes it real, not abstract
4. Have students solve problems with their actual calculators
5. Show the consequences: "If we used 26 AWG for motor wire, it would literally catch fire. Let me show you the math..."

11.2 For Discord/Asynchronous

1. Post this guide as a PDF in OneDrive
2. Create a "Circuit Math Quiz" with 5 problems
3. Students submit answers → you review → clarify misunderstandings
4. Reference specific sections: "Check Section 4 about battery capacity"

11.3 Common Mistakes to Address

- Forgetting to convert mA to A (always convert units first!)
- Mixing up series vs parallel rules
- Not accounting for efficiency losses in BECs
- Using 100% battery capacity instead of 80% safe limit

11.4 Motivation

"We're not just building a toy plane. We're doing REAL aerospace electrical engineering. These are the EXACT same calculations that Boeing uses for 737s, SpaceX uses for rockets, and NASA uses for Mars rovers. You're learning professional-level skills right now."

Conclusion

Use this guide to teach your team the fundamentals they need to understand WHY we made our design decisions, not just HOW to connect wires.

Good luck!